

Product Discontinuation Analysis in the Cosmetics Industry

1. Introduction

The beauty industry is a multi-billion-dollar industry growing exponentially with thousands of new products introduced every year. Safety issues of products and obsolescence risks, however, are very threatening to companies and consumers alike. Cosmetic products tend to contain potentially hazardous chemicals in their formulas, which attract regulatory scrutiny, product recall, and the risk of lawsuits. Such risks not only hurt brand reputation but also result in loss of money and consumer confidence. One of the most cited cases is that of Johnson & Johnson, who were taken to court for toxic chemicals in their products, underlining the need for proactive risk assessment in the industry. This study addresses two significant research questions:

1. Can we classify products based on their likelihood of discontinuation? Machine learning models such as Logistic Regression and Random Forest will be used to forecast product discontinuation and identify high-risk products beforehand.
2. Detecting trends found in cosmetic formulations among different brands. The Clustering algorithms, such as K-means and Hierarchical Clustering, will be employed to segment brands based on usage patterns of chemicals. This way, similarities in formularies can be determined, market trends can be established, and brand strategy differences in cosmetic formulation can be uncovered. These can assist companies in comparative benchmarking with others and data-driven product innovation strategy-formulation.

To achieve these objectives, data preprocessing procedures will be conducted, including date field conversion, missing value treatment, categorical variable encoding, and data aggregation at the product level. Preliminary findings indicate that chemical number and some product categories significantly impact discontinuation patterns.

2. Data

2.1. Data Setting

The data used for this project come from an open-source cosmetic product analysis database that includes information regarding chemical composition, types of products, and the history of discontinued products. Key characteristics include Product Name, Main Category, Subcategory, Chemical Name, Chemical ID, Count of Chemicals, and Date Discontinued. The purpose of this

analysis is to examine whether chemical composition can relate to whether the product has been discontinued, along with factors contributing to a product's shelf life on the market.

Before performing any analysis, preprocessing of the data was performed for consistency and accuracy. The following were the principal steps taken: Handling Missing Values: Categorical features (e.g., BrandName, CasNumber) were replaced with "Unknown". Date fields (InitialDateReported, MostRecentDateReported) were standardized to a date format for time-based analysis. High missing value columns (CSFId, CSF) were dropped. Feature Engineering: Extracted year values from date columns (InitialYear, MostRecentYear) for better temporal analysis. ☐ Computed Product_Lifespan as the difference between MostRecentDateReported and InitialDateReported. Data Transformation: DiscontinuedDate column was converted into a binary variable (Discontinued) indicating if a product was discontinued (1) or not (0). ☐ One-hot encoding was applied to categorical variables (PrimaryCategory, SubCategory, ChemicalName) to prepare them for modeling. ☐ Normalization of numerical variables (ChemicalCount, Product_Lifespan) was performed to scale them for better model performance. Data Aggregation & Duplicates Handling: ☐ The data were sorted at the formulation level of product so that each record is a unique combination of product, category, and chemical composition.

2.2. Data Description

Summary statistics reveal key insights into the dataset. For instance, some brands have as few as 1 product while others have thousands. Most products contain 1-2 chemicals with a maximum of 9. This can be observed in Table 1.

Table 1. Data Descriptive Statistics

Variable	Min	Max	Mean	Median	Std Dev
Chemical Count	1	15	3.8	3	2.1

Products	1	2,500	178	50	450
Per brand					
Chemicals	1	500	35	12	78
Per category					

A boxplot (Figure 1) illustrates the distribution of chemical count across all categories, highlighting variations. The plot reveals that most products contain a low number of chemicals, with a few categories displaying a wider range. Baby Products and Personal Care Products consistently have lower chemical counts, likely due to stricter safety regulations.

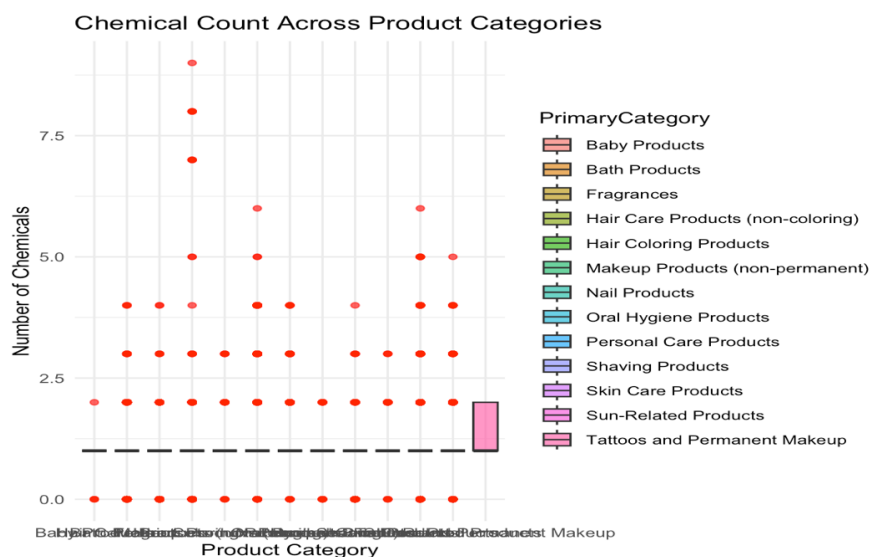


Figure 1. Boxplot of Chemical Count Across Product Categories

3. Empirical Analysis

3.1. Model

We have used both Logistic Regression and Random Forest models. Logistic regression was used as it provided interpretable insights into the factors affecting discontinuation and as it is very useful for Binary classification problems to understand variable importance. The model equation for Logistic regression used was `logit_model <- glm(Discontinued ~ ., data = train_data, family = binomial)`. Random Forest was chosen for its ability to capture complex interactions between variables and its ability to handle missing data. It also provides feature importance rankings to help

identify key factors affecting discontinuation. The model formula used for Random Forest was `rf_model <- randomForest(factor(Discontinued) ~ ., data = train_data_encoded, ntree = 100)`. We have used K-Means Clustering to analyze trends in cosmetic formulations across brands. K-Means was chosen since it clusters brands based on chemical usage patterns well, allowing us to find formulation trends. It can detect groups of brands that share a similar mix of ingredients, and this can contribute to information regarding market positioning and product development.

3.2. Estimation Results

The model's results indicate that the most influential factors affecting discontinuation were that some product categories were inherently more likely to be discontinued, more information is provided in Figure 2A. The main categories influencing discontinuation is given in Figure 2B. Personal Care Products, Baby Products, and Hair Care Products exhibit the highest discontinuation risks. Nail Products and Permanent Makeup are the most stable categories, showing low discontinuation rates. Regulatory trends, ingredient availability, and market demand shifts influence discontinuation patterns

PrimaryCategory <fct>	DiscontinuationRate <dbl>
1 Personal Care Products	0.344
2 Baby Products	0.224
3 Hair Care Products (non-coloring)	0.210
4 Hair Coloring Products	0.209
5 Fragrances	0.191
6 Sun-Related Products	0.173
7 Bath Products	0.160
8 Skin Care Products	0.149
9 Shaving Products	0.149
10 Makeup Products (non-permanent)	0.110
11 Oral Hygiene Products	0.0514
12 Nail Products	0.0498
13 Tattoos and Permanent Makeup	0.00677

Figure 2A. Categories present in Primary category that affect Discontinuation

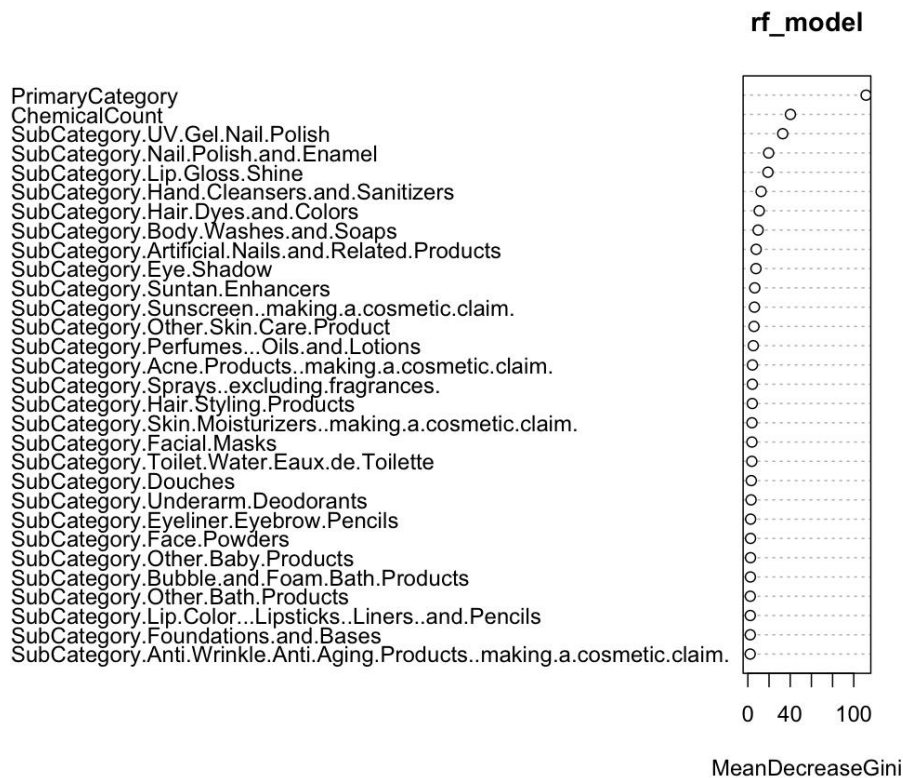


Figure 2B. Categories present in Primary category that affect Discontinuation

Products with more ingredients and higher chemical count have a higher chance of discontinuation. Few product types such as UV Gel nail Polish, Lip Gloss, Hair Dyes show strong correlation with discontinuation. Analysis of the clustering identified separate groups of brands based on chemical usages. Figure 2C shows more information on this.

ChemicalCount_Bins	DiscontinuationRate
<fct>	<dbl>
1 6-10	0.437
2 3-5	0.132
3 0-2	0.111

Figure 2C. Chemical count of products that relate to discontinuation

Cluster 2 (Red/Orange) contains primarily premium brands, such as Charlotte Tilbury, NYX, SEPHORA, and NARS, which are characterized by high ingredient diversity, thereby creating complex formulations that could lead to potential discontinuation risks. In contrast, Cluster 1 (Green) consists of brands like Palladio, No7, CoverGirl, and The Body Shop, which are using

simpler formulations that ultimately contribute to a longer product lifecycle. Business implications are the formulation optimization, supply chain risk mitigation, and marketing strategy outlining. This gives high-ingredient brands an opportunity to promote their products based into luxury, while the low-ingredient ones can market theirs in line with clean beauty, therefore enhancing their appeal. Figure 2D has more information.

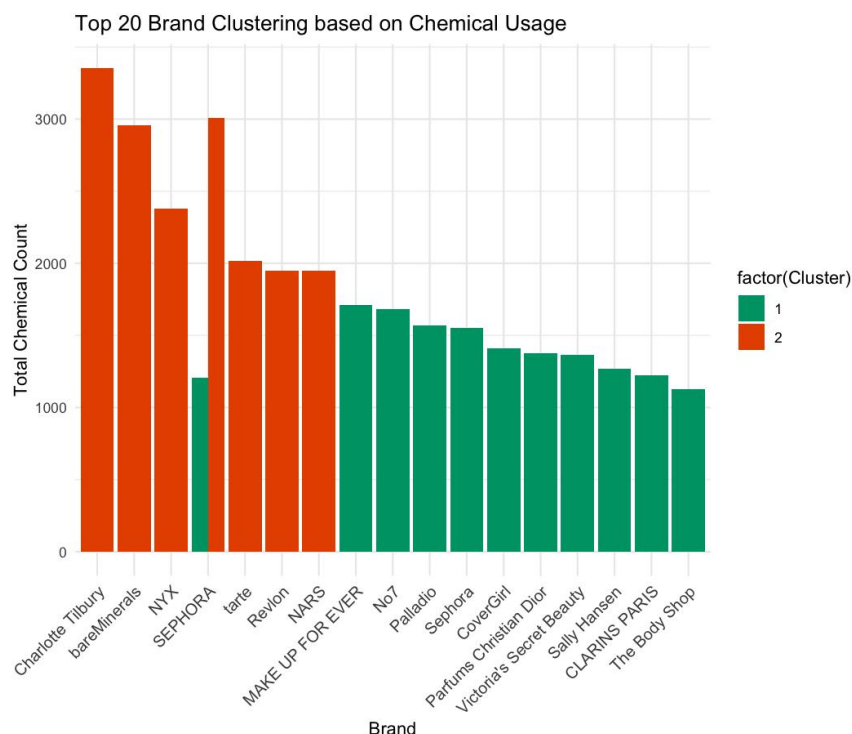


Figure 2D. Clustering graph

Figure 2E and Figure 2F contain the accuracy of the models followed with the confusion matrix.

```
> confusionMatrix(rf_pred, factor(test_data$Discontinued))
Confusion Matrix and Statistics
```

	Reference	
Prediction	0	1
0	6992	1025
1	1	1

Accuracy : 0.8721
95% CI : (0.8645, 0.8793)

Figure 3E. Confusion matrix for Random Forest

```
> confusionMatrix(factor(logit_pred_class), factor(test_data$Discontinued))
Confusion Matrix and Statistics

          Reference
Prediction  0      1
0      6989 1023
1         4     3

      Accuracy : 0.8719
      95% CI   : (0.8644, 0.8792)
```

Figure 4E. Confusion matrix for Logistic regression

4. Conclusions

This analysis identified key factors influencing product discontinuation in the cosmetics industry by applying Logistic Regression and Random Forest models. The moderate predictive power of Logistic Regression failed to perform well in the Random Forest model primarily because both models are tested on imbalanced classes and complex categorical criteria. Primary Category, Chemical Count, and Subcategory were the strong pillars that ruled the longevity of the product. The products are Nail Products, Hair Dyes, and Sunscreens, which seem to have a high risk of being discontinued due to regulatory challenges, ingredient sensitivity, and changing consumers' preferences.

This will provide suitable strategies for product development, marketing, and even supply chain management. This information for the companies will help them optimize formulation, improve stability, and align product offerings with market trends and timely identification of high-risk products to change strategies that would reduce the actual or perceived likelihood of a high rate of discontinuation, increase brand loyalty, and maintain competition.

5. Limitations and Future Research

This Challenges in the Analysis:

1. ☐ Imbalanced Dataset-Most of the active products made it hard to predict discontinuations.

Solution: Apply SMOTE or collect more balanced data.

2. □ □ Absence of Market Trends-Data on sales, pricing, or consumer demand, which affect discontinuation, are not available.

Solution: Include social media sentiment, competitor trends, and sales data.

3. □ □ Limits of Feature Engineering-Chemical Names were dropped, losing valuable insights on the active ingredient.

Solution: Perform text mining of ingredients to analyze the ingredient's influence.

Future work:

1. Testing of more advanced models, such as XGBoost & Neural Networks.
2. Increase predictions by PCA & feature selection.
3. Investigate historic trends & regulations for adequacy of forecasting.
4. Develop a real-time tool for early risk detection.

References

1. Time. (n.d.). *Johnson & Johnson is just the tip of the toxic iceberg*. Time Magazine. <https://time.com/4239561/johnson-and-johnson-toxic-ingredients/>
2. Data.gov. (2024). *Chemicals in cosmetics dataset*. U.S. Government Open Data. <https://catalog.data.gov/dataset/chemicals-in-cosmetics-d55bf>

Cover page

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I am Likith Reddy Challa, a Graduate student pursuing Master's in Data Analytics Engineering at George Mason University. I have experience with analytical tools such as SQL, Power Bi, Excel, R, Python and have worked in a data analytics project during my undergraduate studies. I like to watch movies and play video games where I have helped host tournaments. I am excited to learn more and work towards my current field of data analytics.

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I am Sai Praneet Reddy Chinthala, a Master's student in Data Analytics Engineering at George Mason University. My background lies in machine learning, big data, and data analytics using skills like Python, R, and SQL to get the required insights. I also worked in predictive analytics, ETL operations, and AI automation. In my spare time, I am an avid player of cricket and have won several tournaments, while I have managed to organize various university events effectively. My diverse experience has enhanced my problem-solving skills, adaptability, and leadership, which have ignited my passion for lifelong learning and innovative problem-solving.

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I am Medha Chada, current Master's student in Data Analytics Engineering at George Mason University with a strong machine learning, statistical modeling, and data visualization foundation. Proficient in programming languages like Python, R, and SQL, I also have some experience with AWS tools like Glue and EC2. Hands-on experience working with ETL processes, predictive analytics, and data-driven decision making. In my undergraduate years, I was much involved in event management and was also a member of the dance club, thus combining the creative and organizational aspects of things.

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I am Sai Jayanth Kona, a Master's student in Data Analytics Engineering at George Mason University. I have a background in IT with a Bachelor's degree from Osmania University. I gained hands-on experience as a Front-End Developer Intern at Seneca Global, improving my web development skills. I am skilled in Python, have basic knowledge of R, and understand DBMS and SQL for managing databases. I enjoy problem-solving and continuously learning. I am eager to apply my skills in data engineering and analytics and look forward to new opportunities to grow in this field.